

LESSON 3-6 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Customary Unit Conversions**

Maryland Standard 6.RP.A.3d · Learning target – I can convert between customary units by multiplying or dividing by the conversion rate.

REMEMBER

- $12 \text{ in} = 1 \text{ ft}$ · $3 \text{ ft} = 1 \text{ yd}$ · $16 \text{ oz} = 1 \text{ lb}$ · $2 \text{ cups} = 1 \text{ pt}$ · $2 \text{ pt} = 1 \text{ qt}$ · $4 \text{ qt} = 1 \text{ gal}$
- From a larger unit to a smaller unit, **multiply**. From a smaller unit to a larger unit, **divide**.
- Check: more of a small unit, fewer of a large unit.

WORKED EXAMPLE

A banner is 54 inches long. How many feet long is it?

Step 1 $1 \text{ ft} = 12 \text{ in}$. Inches to feet is small to large, so divide.

Step 2 $54 \div 12 = 4.5$

Answer 4.5 feet (4 feet 6 inches)

WORD BANK

convert – write the same amount in a new unit

multiply – use it when the new unit is smaller

divide – use it when the new unit is bigger

facts – $1 \text{ ft} = 12 \text{ in}$ · $1 \text{ yd} = 3 \text{ ft}$ · $1 \text{ lb} = 16 \text{ oz}$ · $1 \text{ gal} = 4 \text{ qt}$ · $1 \text{ qt} = 2 \text{ pt}$ · $1 \text{ pt} = 2 \text{ cups}$

PRACTICE

show your work · label units

1. Convert each measure.

Step 1 Is the new unit smaller? Multiply by the fact.

Step 2 Is the new unit bigger? Divide by the fact.

a. $8 \text{ ft} = \underline{\hspace{1cm}} \text{ in}$

b. $21 \text{ ft} = \underline{\hspace{1cm}} \text{ yd}$

c. $5 \text{ lb} = \underline{\hspace{1cm}} \text{ oz}$

d. $12 \text{ qt} = \underline{\hspace{1cm}} \text{ gal}$

e. $7 \text{ pt} = \underline{\hspace{1cm}} \text{ cups}$

f. $40 \text{ oz} = \underline{\hspace{1cm}} \text{ lb}$

2. Write $<$, $>$, or $=$ in each blank.

a. $3 \text{ ft} \underline{\hspace{1cm}} 40 \text{ in}$

b. $5 \text{ pt} \underline{\hspace{1cm}} 3 \text{ qt}$

c. $2 \text{ lb} \underline{\hspace{1cm}} 30 \text{ oz}$

d. $9 \text{ ft} \underline{\hspace{1cm}} 3 \text{ yd}$

3. Complete the ratio table for gallons and quarts.

Gallons	1	3		7.5
Quarts	4		20	

- 4. Punch Bowl.** A punch bowl holds 3 gallons. How many 1-cup servings does it hold? (1 gal = 4 qt, 1 qt = 2 pt, 1 pt = 2 cups)

Step 1 Gallons to quarts: $\times 4$.

Step 2 Quarts to pints: $\times 2$.

Step 3 Pints to cups: $\times 2$.

Show your work.

Answer _____ cups

- 5. Reasoning.** Tasha needs 2 yards of ribbon. She has 70 inches. She says she has enough because 70 is greater than 2. Is she right? Explain.

Frame Tasha is ____ because 2 yards = ____ inches, and 70 inches is ____.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____

AT HOME Find a container at home labeled in quarts or ounces. Convert its size to another unit.

LESSON 3-6 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Metric Unit Conversions**

Maryland Standard 6.RP.A.3d • Learning target – I can convert between metric units by multiplying or dividing by powers of ten.

REMEMBER

- $1 \text{ km} = 1,000 \text{ m}$ • $1 \text{ m} = 100 \text{ cm}$ • $1 \text{ cm} = 10 \text{ mm}$ •
 $1 \text{ kg} = 1,000 \text{ g}$ • $1 \text{ L} = 1,000 \text{ mL}$
- Larger unit to smaller unit: **multiply**. Smaller to larger: **divide**. Move the decimal point.

WORKED EXAMPLE

A bag of apples has a mass of 2,400 g. What is its mass in kilograms?

Step 1 $1 \text{ kg} = 1,000 \text{ g}$. Grams to kilograms is small to large, so divide.

Step 2 $2,400 \div 1,000 = 2.4$ (move the decimal point 3 places left)

Answer **2.4 kg**

WORD BANK

kilo- – 1,000 of the unit

centi- – one hundredth of the unit

milli- – one thousandth of the unit

match units – change to the same unit before you divide

PRACTICE

show your work • label units

1. Convert each measure.

a. $3.5 \text{ km} = \underline{\hspace{2cm}} \text{ m}$

b. $85 \text{ cm} = \underline{\hspace{2cm}} \text{ m}$

c. $6 \text{ L} = \underline{\hspace{2cm}} \text{ mL}$

d. $450 \text{ g} = \underline{\hspace{2cm}} \text{ kg}$

2. **Water Station.** A cooler holds 18 L of water. Each cup holds 300 mL. ($1 \text{ L} = 1,000 \text{ mL}$)

Step 1 Change liters to milliliters: $\times 1,000$.

Step 2 Divide by 300 mL for each cup.

a. How many milliliters are in the cooler? mL

b. How many cups can be filled? cups

3. **Relay Race.** Four runners each run 800 m. How many kilometers do they run in all?

Step 1 Multiply 4×800 to get meters.

Step 2 Divide by 1,000 to get kilometers.

a. Total meters: m

b. Total kilometers: km

4. **Baking.** A recipe needs 1.5 kg of flour. Hana has 3 bags with 400 g of flour in each.

Step 1 Find Hana's grams: 3×400 .

Step 2 Change 1.5 kg to grams: $\times 1,000$.

Step 3 Compare the two amounts.

- a. How many grams of flour does Hana have? _____ g
b. Does she have enough? How many grams more or less? _____

5. **Find the Error.** Ian wrote $4.5 \text{ kg} = 0.0045 \text{ g}$. Explain his mistake and find the correct mass.

Frame Ian's mistake is _____. The correct mass is _____ g because _____.

TEST PRACTICE

choose the best answer

6. A pencil is 16 cm long. How many millimeters long is it?

A 1.6 mm **B** 160 mm **C** 1,600 mm **D** 0.16 mm

7. Which measure is the same as 0.6 L?

A 6 mL **B** 60 mL **C** 600 mL **D** 6,000 mL

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____

AT HOME Find a food label with grams or milliliters. Convert the amount to kilograms or liters.